Vegetable juices, root-vegetable-containing: HMTc Standard

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CORRESPONDENCE

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for vegetable juices, root--vegetable-containing in Beverages, evaluated on ready-to-drink liquid, as consumed (wet weight, single strength); a concentrate or a powder-to-liquid product is assessed as reconstituted to single strength per the manufacturer's own directions, which is the basis EU 2023/915 and Codex CXS 193 write into their juice cells and Hong Kong Cap 132V Note 11 writes into its beverage entries. 5 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum level for this exact beverage form and analyte in the row's declared native basis, with disclosed read-across where the exact cell is unregulated, a substantiated not-material determination where a species is bounded or is a method artifact, and a control where no defensible sovereign number and no supportable near-neighbour exist. Never occurrence-derived. Ready-to-drink and dry-commodity bases are never mixed and no brewing, dilution or reconstitution factor is invented.

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS

Lead (Pb)	Tier 1	30	human-grade read-across (European Union)
Cadmium (Cd)	Tier 1	100	human-grade read-across ()
Total arsenic (tAs)	Tier 2	500	screening ceiling (Hong Kong)
Inorganic arsenic (iAs)	Tier 1	20	human-grade read-across (European Union)
Mercury, total (tHg)	Tier 1	10	human-grade read-across (Hong Kong)
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	250	government maximum level
Tin (Sn)	Tier 2	100,000	government maximum level
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	500	screening ceiling (Hong Kong)
Hexavalent chromium (Cr-VI)	Tier 2	n/m	not material in this matrix

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	30	European Union — human-food analog
Cadmium (Cd)	100	— human-food analog
Total arsenic (tAs)	500	government maximum level
Inorganic arsenic (iAs)	20	European Union — human-food analog
Mercury, total (tHg)	10	Hong Kong — human-food analog

Methylmercury (MeHg)	n/m	Methylmercury has no formation pathway in a root-vegetable--containing juice: there is no marine bioaccumulation step in the supply chain and no anaerobic, sulfate-reducing methylation environment in the product or its process. Methylmercury is a component of total mercury (MeHg <= tHg) and is governed by and bounded by the total-mercury cell; no separate methylmercury ceiling is published.
Nickel (Ni)	250	government maximum level
Tin (Sn)	100,000	government maximum level
Aluminium (Al)	control	No credible sovereign sets an aluminium maximum level for a root-vegetable-containing juice. The only sovereign aluminium values in a dietary basis are the packaged-drinking-water quality standards (US 21 CFR 165.110 secondary allowable level 200 ug/kg, EU 2020/2184, Health Canada 2900 ug/kg), which govern water sold as water and are Category 8 instruments, and the FSANZ infant-formula aluminium limits, which are Category 1 and are excluded from this category's anchor set by the taxonomy boundary rule. In this form aluminium is a packaging and processing signal (aluminium cans and can linings, closures, filter aids, additive carry-over) rather than a raw-material contaminant.
Total chromium (Cr)	500	government maximum level

Hexavalent chromium (Cr-VI)	n/m	Hexavalent chromium is not stable in a root-vegetable-containing juice. At the pH of this product form (typically pH 3.8 to 4.8 after the acidification that shelf-stable root-vegetable juices require, with organic acids present) Cr(VI) is reduced to Cr(III) by the organic acids, sugars, ascorbate and polyphenols present in the matrix, so a positive Cr(VI) result is an artifact of the total-chromium determination rather than a species present in the beverage as consumed. The reference-method basis for this determination is the in-house species arbitration at data/workbench/standards/-2026-07-31-chromium-formula--adjudication.md, which found that the superseded ion-exchange plus electrothermal-AAS methods that generated the historical Cr(VI) values in liquid food matrices are not reproduced by the modern reference method: saraiva2021 (species-specific isotope dilution HPLC-ICP-MS, limit of quantification 0.049 ug/kg) quantified no Cr(VI) in any milk or formula sample while resolving Cr(III) and total chromium at single-digit ug/kg. That adjudication was drawn on a near-neutral aqueous food matrix; the acidic beverage matrices in this row are a more strongly reducing environment still, so the artifact conclusion carries a fortiori. The health-relevant species is controlled through the total-chromium cell and the reflex rule below, and this determination is a method finding, not an occurrence-derived value: no number is taken from it.

4 | Cite this standard

Pendergrass, K. "Vegetable juices, root-vegetable-containing." In Heavy Metal Tested & Certified (HMTc) Beverages standard, version 1.0 (2026-08-17). Institute of Contaminant Standards (ICS), 2026.

References

[1] European Union — human-food maximum levels
Binds Pb, iAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] Hong Kong — human-food maximum levels
Binds tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/-vegetable-juices-root-vegetable-containing. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-vegetable-juices-root-vegetable-containing.